

SIT723 Research Techniques & Applications / SIT792 Minor Thesis**Task 3.4D: Research Methodology**

Measuring Semantic Stability of Interpretation in Large Language Models: A UMLS-Grounded Semantic Entropy Framework for Clinical NLP Evaluation

Supervisor: Bahadorreza Ofoghi | Deakin University | Draft for Supervisor Review

1. Introduction and Research Context

Large language models (LLMs) have demonstrated strong performance across a wide range of natural language processing tasks, including biomedical named entity recognition, clinical question answering, and medical report generation. However, high benchmark accuracy does not necessarily indicate that a model produces stable semantic interpretations. A model may generate different conceptual outputs when the same clinical meaning is expressed through different but semantically equivalent phrasings -- a form of unreliability that current evaluation frameworks are not designed to detect.

This research addresses that gap by proposing and empirically validating a UMLS-grounded semantic entropy framework for measuring the stability of semantic interpretation in clinical NLP. Semantic entropy, introduced by Kuhn et al. (2023) and validated by Farquhar et al. (2024) for hallucination detection, measures uncertainty at the level of meaning rather than token probability. The key innovation of the present study is replacing the approximate NLI-based output clustering used in prior work with a formal UMLS Concept Unique Identifier (CUI) equivalence criterion, enabling ontology-grounded stability measurement in the biomedical domain where formal concept systems exist.

The study is grounded in four research gaps identified in the critical literature review (Task 2.3D): (1) overreliance on accuracy-based evaluation that can hide semantic instability under meaning-preserving variation; (2) absence of a UMLS-grounded semantic entropy framework for clinical NLP evaluation; (3) unknown effect of domain-specific biomedical pretraining on semantic stability under controlled perturbation; and (4) lack of a structured, ontology-grounded metric for evaluating the illusion of understanding in clinical language settings.

2. Research Questions

The following three research questions are expanded from Task 3.1P. Each question is directly motivated by one or more of the four research gaps identified in the literature review and is operationalised through pre-specified statistical tests, pre-registered on OSF before any data collection.

2.1 RQ1: Characterisation of Semantic Instability

Which linguistic properties of meaning-preserving perturbations — perturbation type, magnitude of lexical change, and grammatical category of modified tokens — are the strongest independent predictors of semantic entropy in clinical LLM outputs, and do these predictors interact?

Motivation and gap addressed

This question addresses Gap 2 (Task 2.3D): semantic entropy has not been fully grounded in biomedical concept systems such as UMLS, and no study has systematically examined which linguistic properties of meaning-preserving perturbations drive instability in clinical language settings. Prior semantic entropy work (Kuhn et al., 2023; Farquhar et al., 2024) treats entropy as a single aggregate measure without decomposing it by linguistic dimension. This question fills that gap by using UMLS CUI clustering as the formal equivalence criterion and attributing entropy to specific input properties through a pre-registered mixed-effects regression.

Sub-questions

- How does semantic entropy vary across perturbation types (back-translation, controlled paraphrase, synonym substitution, and syntactic reordering) when all other linguistic factors are held constant?
- Does the magnitude of lexical change, measured as normalised Levenshtein edit distance, independently predict semantic entropy after controlling for perturbation type and grammatical category?
- Do different grammatical categories (nouns, verbs, modifiers, function words) contribute differently to semantic entropy, and does a perturbation-type by grammatical-category interaction exist?

Statistical operationalisation

Primary analysis: Mixed-effects regression model.

Model specification: $\hat{H}(x_0) \sim \text{PerturbationType} \times \text{LinguisticCategory} + \text{Magnitude} + \text{Accuracy} + (1|\text{Instance}) + (1|\text{Model})$

Estimation: Restricted Maximum Likelihood (REML). Pre-specified minimum effect size: $f^2 \geq 0.04$ for fixed effects of interest at BH-FDR $q = 0.05$. Marginal and conditional R^2 reported. Fixed-effect 95% confidence intervals via parametric bootstrap (1,000 iterations).

Expected deliverables

- Semantic entropy scores broken down by perturbation type, lexical change magnitude, and grammatical category of modified tokens.
- Mixed-effects regression results showing independent effect sizes (f^2) for each linguistic predictor with 95% bootstrap confidence intervals.
- Test of the perturbation-type by grammatical-category interaction term and its practical significance.
- Identification of which linguistic factor most strongly drives semantic instability in clinical concept normalisation.

2.2 RQ2: Accuracy-Stability Dissociation

Does semantic entropy detect systematic interpretation instability in high-accuracy LLM outputs that traditional evaluation metrics fail to reveal, and does this accuracy–stability dissociation hold across both encoder-only and generative model architectures?

Motivation and gaps addressed

This question addresses Gaps 1 and 4 (Task 2.3D). Gap 1 identifies the overreliance on benchmark accuracy despite evidence that accuracy can hide instability under meaning-preserving variation (Sclar et al., 2024; Raj et al., 2023; Hager et al., 2024). Gap 4 identifies the absence of a structured, ontology-grounded metric for evaluating the illusion

of understanding in clinical language settings. This question tests whether a model performing well on accuracy can simultaneously exhibit elevated semantic entropy $\hat{H}(x_0)$ in the high-accuracy quartile, demonstrating that accuracy-based evaluation systematically underestimates reliability risk.

Sub-questions

- Can a model achieve high task accuracy while still producing inconsistent semantic outputs across meaning-equivalent inputs — specifically in the high-accuracy quartile of the output distribution?
- Does the accuracy–stability dissociation appear in both encoder-only models (CUI-ranking entropy) and generative models (text-generation entropy), or is it architecture-specific?
- How often do apparently correct outputs remain semantically unstable when measured through UMLS-grounded semantic entropy rather than surface accuracy alone?

Statistical operationalisation

Primary analysis: One-tailed Wilcoxon signed-rank test comparing $\hat{H}(x_0)$ in top-accuracy quartile vs. bottom-accuracy quartile per model. Pre-specified threshold: rank-biserial $r \geq 0.30$. Global dissociation test: Spearman rank correlation between per-instance accuracy and $\hat{H}(x_0)$, pooled via Fisher's z-transformation; pre-specified threshold $|\rho| < 0.30$.

Expected deliverables

- Comparison of task accuracy and normalised semantic entropy $\hat{H}(x_0)$ across the same outputs, per model.
- Statistical evidence of whether high-accuracy instances show significantly elevated semantic entropy (Wilcoxon signed-rank, one-tailed, rank-biserial $r \geq 0.30$).
- Separate analysis for encoder-only (CUI-ranking entropy) and generative (text-generation entropy) architectures.
- Results demonstrating whether semantic entropy provides reliability information that accuracy metrics cannot detect.

2.3 RQ3: Domain Adaptation and Semantic Stability

Does domain-specific biomedical pretraining reduce semantic entropy relative to general-purpose models at equivalent parameter scale, and does this stability advantage generalise from medical concept normalisation to biomedical and general-domain question answering?

Motivation and gap addressed

This question addresses Gap 3 (Task 2.3D): biomedical LLMs have not been systematically compared in terms of semantic stability under controlled meaning-preserving perturbation. While the biomedical NLP literature establishes that domain-adapted models such as BioBERT (Lee et al., 2020), PubMedBERT (Gu et al., 2021), and BioMistral (Labrak et al., 2024) outperform general-purpose models on task accuracy, it is unknown whether any accuracy advantage extends to semantic consistency. The within-scale comparison design isolates domain-specific pretraining from the confound of model scale.

Sub-questions

- Do biomedical models show lower semantic entropy than general-purpose models of equivalent scale in within-scale comparisons: BioBERT vs. BERT-base (both 110M parameters) and BioMistral-7B vs. FLAN-T5-XXL (large-scale tier)?
- Does any stability advantage observed on MedMentions (medical concept normalisation) persist on BioASQ (biomedical QA) and attenuate on SQuAD 2.0 (general-domain QA), indicating domain-specific rather than fully general stability benefits?
- Does domain adaptation improve the consistency of semantic interpretation under meaning-preserving variation, or does it only improve benchmark accuracy?

Statistical operationalisation

Primary analysis: One-tailed Mann-Whitney U tests in two pre-specified within-scale pairs. Pair 1: BioBERT vs. BERT-base (both 110M). Pair 2: BioMistral-7B vs. FLAN-T5-XXL (large-scale). Pre-specified threshold: rank-biserial $r \geq 0.30$ in each pair. Generalisation condition: attenuation of advantage on SQuAD 2.0 relative to MedMentions, tested by comparing effect sizes across datasets. BH-FDR correction at $q = 0.05$ across all confirmatory tests.

Expected deliverables

- Within-scale semantic entropy comparisons: BioBERT vs. BERT-base (110M) and BioMistral-7B vs. FLAN-T5-XXL (large-scale), with effect sizes and 95% bootstrap confidence intervals.
- Semantic entropy results across all three datasets (MedMentions, BioASQ, SQuAD 2.0) enabling a domain-continuum analysis.
- Statistical evidence of whether domain adaptation improves semantic stability as well as task accuracy.
- Discussion of whether any stability advantage is domain-specific or generalises across task formats, with implications for model selection in clinical NLP.

3. Research Methods

3.1 Overview of Research Methods Considered

The following table provides a succinct overview of the main research methods relevant to the study domain, with their strengths, weaknesses, and suitability for the present research questions.

Method	Description	Strengths	Weaknesses	Suitability for this study
Semantic entropy (UMLS-grounded CUI clustering)	Computes Shannon entropy over meaning-level output clusters defined by UMLS CUI equivalence rather than token probability.	- Formal ontological grounding. - Model-agnostic. - No fine-tuning required. - Directly addresses meaning-level stability.	- Requires UMLS entity linker (noise source). - Only applicable to domains with formal concept systems.	PRIMARY METHOD. Directly operationalises all three RQs. The UMLS extension is the study's main contribution.

Method	Description	Strengths	Weaknesses	Suitability for this study
Accuracy-based benchmarking	Evaluates model performance against ground-truth labels using standard classification metrics.	- Well-established. - Easy to compare across studies. - Interpretable to practitioners.	- Does not capture stability under variation. - Rewards correct-once behaviour.	Used as BASELINE COMPARISON only. RQ2 specifically tests its limitations.
SelfCheckGPT (consistency-based detection)	Samples multiple outputs and checks consistency using NLI, n-gram, or BERTScore signals without formal clustering.	- Zero-resource. - No clustering required. - Multiple signal types.	- No formal equivalence criterion. - Cannot distinguish CUI-level meaning change.	Used as COMPARISON BASELINE to demonstrate the advantage of UMLS-grounded entropy over informal consistency metrics.
Meaning-preserving perturbation testing	Generates semantically equivalent input variants through back-translation, paraphrase, synonym substitution, and syntactic reordering, then measures output stability.	- Controlled exposure of instability. - Isolates linguistic factors. - Supports causal decomposition.	- Perturbation quality depends on automated pipeline. - May not capture all meaning-change dimensions.	CORE METHODOLOGY for generating the input variation that entropy is measured over. Supports RQ1 decomposition.
Calibration analysis (ECE, temperature scaling)	Measures the alignment between model-reported confidence and empirical accuracy across probability bins.	- Well-understood. - Identifies overconfidence. - Can be post-hoc applied.	- Requires output probabilities. - Does not measure semantic consistency. - Cannot detect stability under paraphrase.	BACKGROUND ONLY. Helps distinguish semantic stability from calibration. Not a primary method.
Fine-tuning / domain adaptation	Adapts a pre-trained model to a specific domain by training on in-domain labelled data.	- Can improve task performance. - Reduces generalisation gap.	- Requires labelled data. - May reduce generalisability. - Not zero-shot.	NOT used in this study. All models evaluated in zero-shot setting to maximise generalisability and isolate pretraining effect.

3.2 Selected Research Method and Justification

The study adopts a UMLS-grounded semantic entropy evaluation framework as its primary method, combined with meaning-preserving perturbation generation and a six-gate automated quality validation pipeline. This combination is selected for the following reasons.

Why semantic entropy rather than accuracy-based benchmarking

Accuracy-based benchmarking evaluates whether a model produces the correct output on a single presentation of each input. It cannot detect whether the model consistently maps semantically equivalent inputs to the same interpretation. Semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024) directly measures this form of reliability by computing uncertainty over meaning-level output clusters. The UMLS extension of this method -- using CUI equivalence rather than NLI-based approximate clustering -- provides a formal ontological grounding that is not possible in general-domain settings, making the clinical NLP domain an ideal and methodologically privileged setting for this framework.

Why meaning-preserving perturbations rather than adversarial attacks

Adversarial attacks (e.g., TextFooler, HotFlip) often violate semantic meaning in pursuit of prediction flips. The present study requires strictly meaning-preserving variation to attribute entropy to input rephrasing rather than meaning change. The four perturbation methods (back-translation, controlled paraphrase, synonym substitution, and syntactic reordering) are selected because they cover complementary linguistic dimensions -- lexical, syntactic, and translational -- while all being validatable through the six-gate quality pipeline. The Coltekin et al. (2022) review confirms that no single augmentation method dominates across tasks, justifying a multi-method approach.

Why zero-shot evaluation rather than fine-tuning

All models are evaluated in zero-shot or structured-prompt settings with no task-specific fine-tuning. This decision is motivated by three considerations: (1) zero-shot evaluation maximises generalisability of findings to deployment settings where fine-tuning is not available; (2) it isolates the effect of domain-specific pretraining (the variable of interest in RQ3) from fine-tuning effects; and (3) it enables a clean within-scale comparison between biomedical and general-purpose models at equivalent capacity.

Why mixed-effects regression for RQ1

The mixed-effects regression model is selected for RQ1 because it correctly accounts for the hierarchical structure of the data -- multiple perturbations per instance, multiple instances per model -- through random intercepts, while estimating fixed effects for the linguistic predictors of interest. This is the appropriate design for the repeated-measures structure of the entropy dataset. The model is pre-registered on OSF before any data collection, ensuring confirmatory rather than exploratory framing.

Why non-parametric tests for RQ2 and RQ3

Wilcoxon signed-rank (RQ2) and Mann-Whitney U (RQ3) are selected because normalised semantic entropy $\hat{H}(x_0) \in [0,1]$ is bounded and may not follow a normal distribution, particularly at the tails. Non-parametric tests make no distributional assumptions and are robust to the skewness expected in entropy distributions where many instances may cluster near 0 (stable) or 1 (maximally unstable). Effect sizes (rank-biserial r) are pre-specified to ensure practical significance is distinguished from statistical significance.

4. Research Plan with Timelines

The research is organised into five phases spanning the thesis project period. Each phase has defined inputs, outputs, and dependencies. The timeline is designed to be realistic given the available hardware (2× NVIDIA RTX A4000, 16 GB VRAM each) and estimated computational requirements (26-43 hours wall time for all model inference).

4.1 Phase Overview

Phase	Name	Key Activities	Outputs	Dependencies and Risks
1	Pre-registration and Setup	- OSF pre-registration of H1, H2, H3, SA1-SA6. - Environment setup and dependency pinning. - Threshold calibration on pilot pairs.	- Pre-registration identifier. - Docker container image. - Calibrated six-gate thresholds.	- Must be completed before any data collection. - Risk: OSF submission delay - mitigated by drafting registration document in parallel with this task.
2	Dataset Preparation	- Stratified sampling of all three datasets. - scispaCy UMLS linker validation. - Pilot set reservation and exclusion check.	- Three stratified sample files. - Inclusion/exclusion log. - Pilot QA report.	- Depends on Phase 1 completion. - Risk: Higher-than-expected scispaCy rejection rate -- mitigated by 10% buffer sample.
3	Perturbation Generation and Validation	- Generate k=8 perturbations per instance via 4 methods. - Apply six-gate quality pipeline. - Log rejection rates per gate per method.	- Validated perturbation dataset (~15 GB). - Gate rejection rate report. - Perturbation annotation file (type, magnitude, POS category).	- Depends on Phase 2. Estimated 8-12 hours compute. - Risk: High rejection rate in G6 (entity preservation) for rare UMLS concepts -- up to 3 regeneration attempts per slot.
4	Model Inference and Entropy Computation	- Run all 5 models on 9 inputs per instance. - Apply CUI mapping protocol (5 rules). - Compute $\hat{H}(x_0)$ for all instances and models. - Negative control validation. - MetaMap cross-linker reliability check (10% subsample).	- Entropy dataset (n×5 models). - CUI mapping log with rule applied. - Cross-linker reliability report (kappa target ≥ 0.75). - Negative control results.	- Depends on Phase 3. Estimated 26-43 hours wall time. - Risk: VRAM overflow for FLAN-T5-XXL -- mitigated by INT8 quantisation and sequential GPU-2 allocation. - Risk: Cross-linker kappa < 0.75 -- triggers extended MetaMap reconciliation protocol.
5	Statistical Analysis and Write-Up	- Run H1 (mixed-effects regression), H2 (Wilcoxon), H3 (Mann-Whitney U). - BH-FDR correction across all tests. - Run SA1-SA6 sensitivity analyses. - Draft Results chapter. - Release code and data publicly.	- Confirmed or rejected H1, H2, H3 with effect sizes and CIs. - Sensitivity analysis report. - Results chapter draft. - Public GitHub repository and data release.	- Depends on Phase 4. - Risk: Non-normal entropy residuals in H1 -- pre-specified fallback to permutation test documented in pre-registration. - Risk: Insufficient n after exclusions -- documented as scope limitation.

4.2 Detailed Milestone Timeline

Milestone	Phase	Deliverable and Deadline	SMART Criteria
M1: OSF Pre-registration	1	Complete OSF pre-registration of all H1/H2/H3 confirmatory tests and SA1-SA6. Deadline: Before any data collection begins.	- Specific: Pre-registration covers all three hypotheses and six sensitivity analyses. - Measurable: OSF registration identifier obtained and recorded. - Achievable: Form drafted during Task 3.3C; submission takes 1-2 hours. - Relevant: Required for confirmatory framing. - Time-bound: Before Phase 2 begins.
M2: Dataset Preparation	2	Stratified samples extracted: MedMentions (n=550+55 pilot), SQuAD 2.0 (n=200+20), BioASQ (n=150+15). scispaCy validation passed. Inclusion log complete. Deadline: 1 week after M1.	- Specific: Three dataset files with defined stratification. - Measurable: Exact n confirmed; pilot instances flagged and excluded from main analysis. - Achievable: Datasets publicly available; scispaCy runs on CPU. - Relevant: Data quality is the foundation of all subsequent analysis. - Time-bound: 1 week after M1.
M3: Threshold Calibration	2-3	Six-gate thresholds calibrated on 200 positive (UMLS synonym pairs) and 200 negative (different-CUI pairs). Calibration results documented in supplementary materials. Deadline: Before full perturbation generation.	- Specific: 400 calibration pairs, six thresholds verified. - Measurable: False-admission rate $\leq 10\%$ on negative pairs. - Achievable: Proxy pairs drawn from UMLS Metathesaurus directly. - Relevant: Gate integrity is prerequisite for perturbation quality. - Time-bound: Concurrent with M2.
M4: Perturbation Generation	3	k=8 validated perturbations generated for all instances across three datasets. Perturbation annotation complete (type, magnitude, POS category). Deadline: 2 weeks after M3.	- Specific: 8 perturbations per instance, 4 methods, 6 gates. - Measurable: Rejection rate logged per gate; overall acceptance rate $\geq 90\%$. - Achievable: All perturbation models run on CPU/GPU-1. - Relevant: Perturbation dataset is the core experimental input. - Time-bound: 2 weeks; ~8-12 hours compute.
M5: Negative Control Validation	4	Identical-copy and typographic-only control results verified for all 5 models. Any model with $\hat{H} > 0.10$ on identical-copy excluded with documentation. Deadline: Before main inference run.	- Specific: 60 instances with two control variants per instance. - Measurable: $\hat{H} = 0$ for identical copies; $\hat{H} \approx 0$ for typographic variants. - Achievable: Controls generated programmatically. - Relevant: Validates entropy metric before any main analysis. - Time-bound: 2-3 hours compute after perturbation dataset ready.

Milestone	Phase	Deliverable and Deadline	SMART Criteria
M6: Model Inference	4	All 5 models process all 9 inputs per instance across all 3 datasets. Outputs saved to SSD with full logging. Deadline: 2 weeks after M5.	- Specific: 5 models $\times$ (n=550+200+150) $\times$ 9 inputs = ~40,500 inference calls. - Measurable: All output logs present; no missing predictions. - Achievable: GPU-1 (encoders) and GPU-2 INT8 (generative). - Relevant: Inference is the primary data collection step. - Time-bound: 26-43 hours wall time; 2-week window allows for interruptions.
M7: Entropy Computation and Cross-Linker Check	4	$\hat{H}(x_0)$ computed for all instances and models using 5-rule CUI mapping protocol. MetaMap cross-linker kappa check on 10% subsample (target ≥ 0.75). Deadline: 1 week after M6.	- Specific: Full entropy dataset with bootstrap CIs. - Measurable: Kappa value reported numerically. - Achievable: CUI mapping is rule-based; entropy computation is $O(n)$. - Relevant: Entropy dataset is the input to all statistical analyses. - Time-bound: 1 week.
M8: Confirmatory Statistical Analysis	5	H1 (mixed-effects regression), H2 (Wilcoxon), H3 (Mann-Whitney U) conducted. BH-FDR correction applied. SA1-SA6 sensitivity analyses conducted. Deadline: 2 weeks after M7.	- Specific: Three confirmatory tests plus six sensitivity analyses. - Measurable: Each test produces a p-value, effect size, and 95% CI. - Achievable: All analyses run in Python (statsmodels) or R (lme4, simr). - Relevant: Directly answers H1/H2/H3. - Time-bound: 2 weeks; compute is negligible after entropy dataset is ready.
M9: Results Chapter Draft	5	Complete results chapter submitted to supervisor for review. Tables and figures for H1-H3 tests, sensitivity analyses, negative control, and cross-linker reliability. Deadline: 1 week after M8.	- Specific: One full draft chapter with all tables and figures. - Measurable: Supervisor feedback obtained and documented. - Achievable: Results are mechanically derivable from M8 outputs. - Relevant: Draft is required for thesis submission. - Time-bound: 1 week after M8.
M10: Public Code and Data Release	5	GitHub repository with pinned model checkpoints, fixed seeds (42), Docker image, and full analysis pipeline. Public data repository with perturbation datasets and entropy scores. Deadline: Concurrent with thesis submission.	- Specific: MIT licence; Docker image enables bit-identical reproduction. - Measurable: Repository link reported in thesis; Docker image executes without errors. - Achievable: Code maintained throughout all phases. - Relevant: Required for reproducibility and open science commitments. - Time-bound: Final thesis submission date.

5. Ethical Considerations

5.1 Patient Data Privacy

All three datasets used in this study (MedMentions ST21pv, SQuAD 2.0, and BioASQ Task B Phase A) are publicly available de-identified research datasets that contain no patient identifiable information. MedMentions is derived from PubMed abstracts, which are public domain. SQuAD 2.0 is released under CC BY-SA 4.0. BioASQ is available to registered participants under the challenge data agreement. No private clinical records, electronic health records, patient notes, or personally identifiable medical data are accessed, stored, or processed at any stage of this research.

5.2 Confidentiality and Informed Consent

Confidentiality is maintained throughout the research through exclusive use of de-identified, publicly licensed datasets. No human participants are recruited, surveyed, interviewed, or involved in any experimental procedure. The study involves no interaction with individuals and generates no personally identifiable outputs. Informed consent from individual participants is therefore not required. The datasets used include appropriate consent and licensing mechanisms established by their original creators at the time of dataset construction. No individual informed consent process is needed beyond adherence to dataset licensing agreements already in place.

5.3 Clinical Safety and Hallucination Risk

LLMs evaluated in this study produce outputs that are used solely for research measurement. No model output is used for clinical decision-making, patient diagnosis, treatment recommendation, or any form of patient care. The study is explicitly motivated by the clinical safety concern -- identified by Hager et al. (2024, Nature Medicine) and Sivarajkumar et al. (2025, npj Digital Medicine) -- that high accuracy benchmarks can coexist with semantically unstable outputs. By developing a metric that goes beyond accuracy, the study directly contributes to safer evaluation practices for clinical NLP systems. The negative control condition (identical-copy perturbation) validates the entropy metric before any model comparisons are conducted, ensuring that results are not artefacts of the measurement framework.

5.4 HREC Approval and Regulatory Considerations

No Human Research Ethics Committee (HREC) approval is required for this study. The research does not involve human participants, does not involve deception, does not access identifiable personal data, and does not involve any intervention or interaction with people. All data is drawn from publicly available, de-identified research datasets. This determination is consistent with the Deakin University Human Research Ethics procedure for research involving no risk to human participants. The dataset licensing agreements (CC0 for MedMentions, CC BY-SA 4.0 for SQuAD 2.0, BioASQ challenge agreement) constitute the necessary permissions for use of this data in academic research.

5.5 Reproducibility and Research Integrity

All confirmatory analyses (H1, H2, H3) and sensitivity analyses (SA1-SA6) are pre-registered on OSF before any data collection or model inference. The pre-registration identifier is reported in the thesis. All code is released on GitHub under MIT licence with fixed random seeds (numpy: 42; PyTorch: 42; HuggingFace Transformers: 42) and pinned model revision hashes. A Docker container image with all dependencies pinned to exact versions enables

bit-identical reproduction of all results. Full supplementary data -- perturbation datasets, gate scores, CUI mapping logs, model outputs, and entropy scores -- are released publicly alongside the thesis. Any deviation from the pre-registered protocol is documented as exploratory and clearly distinguished from confirmatory findings in the Results chapter.

5.6 Responsible Use of Artificial Intelligence

All LLM inference is conducted in a controlled research environment where models are treated as objects of empirical analysis, not as deployed decision-support tools. The study does not deploy any model in a clinical setting. The study explicitly acknowledges the risk of LLM overconfidence identified by Kadavath et al. (2022) and the evaluation illusion documented by Sivarajkumar et al. (2025). Findings will be reported with appropriate caveats about the scope of the study and the limits of generalisation. The English-centric scope of the study -- a structural limitation of all three datasets -- is explicitly acknowledged as a scope condition in the research questions and the discussion section.

6. Milestones and Sustainability

6.1 Key Milestones Summary

The ten milestones (M1-M10) defined in Section 4.2 mark the critical stages of the research. The overall structure is designed so that no milestone is contingent on an external dependency that cannot be controlled by the research team. The only external dependencies are OSF submission (M1, resolved by pre-drafting the registration) and BioASQ data access (M2, resolved by advance registration with the BioASQ organisers). All other milestones depend only on data, code, and hardware resources that are already available.

6.2 Risk Register and Contingency Plans

Risk	Likelihood	Impact	Contingency Plan
G6 entity preservation rejection rate > 30% for rare UMLS concepts	Medium	Medium	- Up to 3 regeneration attempts per slot already built into pipeline. - If rate exceeds 30%, extend to 5 attempts for medical synonym substitution only. - Document deviation from pre-registration.
Cross-linker kappa (MetaMap vs. scispaCy) < 0.75	Low	High	- Trigger extended reconciliation: manually review a stratified subsample of 50 disagreement cases. - Report kappa numerically regardless of threshold outcome. - If systematic bias identified, apply rule-based correction and document.
FLAN-T5-XXL VRAM overflow under INT8 quantisation	Low	High	- Fallback: FLAN-T5-XL (3B INT8, ~3.5 GB VRAM) as alternative large-scale general model. - Document model substitution as deviation from pre-registration. - Report sensitivity to model choice in SA4.
Mixed-effects regression residuals	Low	Medium	- Pre-registered fallback: permutation-based mixed-effects test (5,000 permutations).

Risk	Likelihood	Impact	Contingency Plan
severely non-normal (H1)			- Report both parametric and permutation results. - Document as sensitivity deviation.
Dataset n falls below target after quality-gate exclusions	Low	Medium	- 10% buffer sample already built into sampling strategy. - If buffer exhausted, report actual n with power re-calculation. - SA6 (exclusion impact analysis) will verify representativeness of included instances.
GPU hardware failure or extended unavailability	Low	High	- Encoder-only models (BERT-base, BioBERT, PubMedBERT) can run on CPU, extending but not preventing completion. - Google Colab Pro (A100) available as emergency cloud fallback for generative models. - Timeline extended by maximum 2 weeks.

6.3 Sustainability Measures

Open science and data accessibility

All research outputs -- code, datasets, model outputs, and entropy scores -- are released publicly under permissive licences (MIT for code; CC BY 4.0 for data). This ensures that other researchers can reproduce, verify, and build upon the findings without requiring access to the original hardware or institutional resources. The Docker container image ensures that results are reproducible at the bit level, not just conceptually.

Dissemination strategy

The primary target venue for journal publication is the Journal of Biomedical Informatics (JBI), which is the leading venue for work at the intersection of biomedical NLP evaluation and clinical informatics. Secondary target venues include npj Digital Medicine (if the clinical safety framing is the lead contribution) and Bioinformatics (if the UMLS-grounded entropy framework is the lead contribution). The work is also suitable for submission to the Transactions of the Association for Computational Linguistics (ACL) if the computational linguistics community is the primary audience. A conference paper version is planned for submission to ACL, EMNLP, or BioNLP workshops to obtain early peer feedback before journal submission.

Long-term relevance and extension pathways

The UMLS-grounded semantic entropy framework developed in this study has clear extension pathways beyond the present scope. First, the framework is directly applicable to any biomedical NLP evaluation task where UMLS concept systems are relevant, including clinical NER, medical relation extraction, and drug-disease association mining. Second, the perturbation pipeline and six-gate quality framework are reusable components that other researchers can apply to different datasets and models. Third, if domain-specific pretraining is found to confer stability advantages (RQ3 confirmed), this finding has direct practical implications for model selection guidelines in clinical NLP -- a policy-level contribution that extends beyond the academic literature.

Computational efficiency and environmental considerations

The study uses INT8 quantisation for large generative models (FLAN-T5-XXL and BioMistral-7B) to reduce VRAM requirements and energy consumption relative to full-precision inference. All encoder-only models run on GPU-1 in batches of 32, which is near-optimal for the available hardware. The estimated total wall time of 26-43 hours is modest for a study of this scope. Cloud compute fallback (Google Colab Pro) is used only if local hardware is unavailable, minimising unnecessary cloud resource usage. All intermediate results are cached to avoid redundant computation.

6.4 Closing Statement

This research methodology document presents a complete, SMART-compliant research plan for the study of semantic stability in clinical LLMs using a UMLS-grounded semantic entropy framework. The three research questions are specific, testable, and directly address four identified research gaps. The selected methods are justified by reference to the reviewed literature and to the methodological requirements of each research question. The ten-milestone timeline is realistic, includes explicit contingency planning, and is designed to produce reproducible, open-science-compliant outputs. The ethical considerations are comprehensive and confirm that no HREC approval is required. The sustainability measures ensure that the research contributes not only to the immediate thesis but to the long-term advancement of reliable clinical NLP evaluation.